

Virtual Friend — Chrome Extension

Full Project Plan: Concept → Architecture → Development → Testing → Chrome Web Store Production

Project goal: Build a small 2D virtual friend that lives in the corner of Chrome webpages, independently performs everyday activities, watches content with the user, reacts occasionally, and can be instantly hidden with a user-controlled Privacy Mode.

1. Product Definition

- **Core identity:** virtual friend, not a pet. Avoid pet-care mechanics such as feeding, hunger meters, or petting as the central loop.
- **Visual scale:** intended to feel roughly 7–8 cm tall on a typical desktop setup. Render in pixels rather than CSS centimeters; provide a size slider.
- **Placement:** small, unobtrusive overlay, normally near the bottom-right or bottom-left corner.
- **Activities:** idle, sitting, eating, sleeping, showering behind a curtain, watching, looking toward the user, stretching, yawning, and short reactions.
- **Privacy Mode:** a focus feature for the user. It makes the friend disappear and stops its visible/audio activity. It is unrelated to the shower scene.
- **Architecture principle:** keep behavior/decision-making separate from rendering/animation so the artwork or animation technology can be replaced later.

2. V1 Scope

Include in V1	Defer to later versions
One character	Multiple characters
2D frame/sprite animation	Live2D
Idle, sitting, eating, sleeping, shower, watching, looking, reactions	AI conversation
Random autonomous behavior	Voice assistant
Simple time-of-day routine	Cloud memory/account system
Generic HTML5 video awareness	Computer vision
YouTube awareness	Complex relationship simulation
Size/position controls	Multiplayer/social features
Hide + Privacy Mode	Cross-device sync
Local settings with chrome.storage	Backend/server
Chrome Web Store packaging	Paid subscriptions

3. Recommended Technology

Layer	Choice	Reason
Extension format	Chrome Manifest V3	Current Chrome extension architecture.
Language	Vanilla JavaScript	Enough for the first version; minimizes unnecessary dependencies.
UI	HTML + CSS	Popup/settings are simple and lightweight.
Rendering	HTML image element / sprite or Easy to author and replace character artwork.	
Isolation	Shadow DOM	Protects the friend from webpage CSS.
Persistence	chrome.storage.local	Local settings without a backend.
Background	MV3 service worker	Commands, extension-level events, and background coordination.
Optional later	Canvas / Live2D	Only if animation complexity justifies it.

4. Project Structure

```

virtual-friend/
  manifest.json
  background/
    service-worker.js
  content/
    friend.js
    friend.css
    animation-engine.js
    state-machine.js
    site-detector.js
  popup/
    popup.html
    popup.css
    popup.js
  options/
    options.html
    options.css
    options.js
  characters/
    default/
    idle/
    sitting/
    eating/
    sleeping/
    shower/
    watching/
    looking/
    reactions/
  icons/
    16.png
    48.png
    128.png

```

5. Manifest & Extension Bootstrap

- Create manifest.json using Manifest V3.
- Define the extension name, version, description, icons, action popup, options page, content scripts, background service worker, storage permission, and keyboard command.
- Keep permissions minimal. Do not request broad capabilities merely because they might be useful later.
- Start with local loading through chrome://extensions → Developer mode → Load unpacked.
- Establish a versioning scheme such as 0.1.0 for development and 1.0.0 for the first public release.

6. Overlay & Shadow DOM

- Content script creates a single host element on supported pages.
- Attach a Shadow Root and render the friend inside it.
- Use position: fixed so the friend stays in the viewport while the page scrolls.
- Default position: bottom-right with a small margin; support all four corners later.
- Use a high z-index inside the isolated component.
- Make the overlay non-invasive: it should not alter page layout or capture clicks unless the user interacts with the friend.

7. Animation Asset Pipeline

- Create each activity as a sequence of transparent PNG/WebP frames or a sprite sheet.
- Keep character dimensions consistent across every animation.
- Design the artwork around the intended small display size; avoid excessive detail that becomes unreadable at 100–150 px.
- Optimize images for file size. Large uncompressed assets are one of the easiest ways to make an extension unnecessarily heavy.
- Use a common naming convention such as idle_001.webp, idle_002.webp, etc.
- The shower activity should be a scene: curtain, steam/water effects, and a fully obscured character silhouette.

8. Animation Engine

- Input: animation name, frames, FPS, loop mode, completion callback.
- Support looping animations for idle/sitting/sleeping.
- Support one-shot animations for reactions, yawns, looks, etc.
- Allow animation interruption when a higher-priority state occurs.
- Pause the renderer when Privacy Mode is active or the friend is hidden.
- Preload only what is necessary; avoid loading every asset into memory immediately.

9. Behavior / State Machine

The behavior system should decide **what the friend does**; the renderer decides **how that activity looks**.

Example state flow:

IDLE → SITTING / WATCHING / EATING / SLEEPING → LOOKING / REACTION → previous state or IDLE.

Each state should define animation, minimum duration, maximum duration, possible next states, probability, and optional conditions.

- Use randomized time ranges instead of fixed intervals.
- Keep behavior weighted rather than purely random.
- Make special behaviors rare enough that they retain novelty.
- Allow context to modify probabilities: video playing increases WATCHING probability; late night increases SLEEPING probability.

10. Core Activities

Activity	Implementation idea	Priority
Idle	Breathing/blinking/subtle motion	P0
Sitting	Chair/floor sitting with small idle actions	P0
Eating	Food prop + eating loop	P0
Sleeping	Sleep pose + breathing	P0
Shower	Opaque curtain + steam/water + silhouette	P0
Watching	Seated/standing pose facing screen	P0
Looking	Short eye-contact/head-turn animation	P0
Reaction	Short smile, surprise, wave, etc.	P1
Stretch/yawn	One-shot idle interruption	P1

11. Browser & Video Awareness

- First support generic HTML5 video elements.
- Detect playing, paused, ended, and fullscreen states.
- Map video state to behavior rather than forcing the friend to stare at every video continuously.
- Add a site-detector module so YouTube-specific logic remains isolated from generic behavior.
- For YouTube, account for dynamic navigation and changing video elements; do not assume a full page reload occurs between videos.
- Later, add site adapters only when they provide meaningful behavior.

12. Time-of-Day Routine

Time	Behavior bias
Morning	Getting ready, eating, sitting, energetic idle
Afternoon	Sitting, working, eating, watching
Evening	Relaxing, watching, eating
Late night	Sleepy, yawning, sleeping

These are probability biases, not rigid schedules. The friend should still feel autonomous.

13. Interaction Model

- Single click can trigger a small glance or reaction.
- Hover can optionally cause a subtle look toward the cursor.
- Double-click can trigger a rare reaction.
- Avoid making the friend behave like a game UI. Interaction should be occasional and optional.
- Provide an option to disable interaction completely.

14. User Controls

Control	Purpose
Show / Hide	Temporarily remove the friend.
Privacy Mode	Focus mode: make the friend invisible and stop visible/audio activity.
Size	Small / default / large slider; default designed around the intended 7–8 cm feel.
Position	Bottom-left/right, top-left/right.
Opacity	Optional subtle transparency control.
Sounds	Enable/disable and volume.
Animation frequency	Reduce or increase autonomous activity.
Watching behavior	Enable/disable video-aware behavior.

15. Privacy Mode Design

- Privacy Mode is a user focus feature, not a character privacy feature.
- When ON, remove/hide the visible friend immediately.
- Stop animation rendering, sounds, and interaction.
- Persist the setting using `chrome.storage.local`.
- Add a keyboard shortcut such as `Ctrl+Shift+P`.
- Optionally provide timed hiding later: 5 minutes, 30 minutes, 1 hour, or until manually restored.
- Keep the underlying state/routine separate from visibility so the friend can naturally resume when shown again.

16. Settings & Persistence

- Centralize defaults in one settings object.
- Store settings in `chrome.storage.local`.
- Suggested settings: `enabled`, `privacyMode`, `size`, `position`, `opacity`, `soundsEnabled`, `volume`, `interactionEnabled`, `videoAwarenessEnabled`, `animationFrequency`.
- Validate settings when loading so corrupted or outdated values fall back to safe defaults.
- Version settings if future releases change the data structure.

17. Performance Requirements

- Avoid high-frequency polling. Prefer DOM/video events, timers, and observers.
- Pause animation when hidden or Privacy Mode is active.
- Avoid decoding dozens of huge frames simultaneously.
- Keep the service worker lightweight; do not use it for continuous animation.
- Measure CPU usage while playing a 1080p/4K video and while the browser is idle.
- Target a companion that feels visually present without becoming a noticeable resource drain.

18. Error & Edge-Case Handling

- If an animation asset is missing, fall back to idle rather than breaking the overlay.
- If the website changes its DOM, recreate the host only if necessary.
- Avoid duplicate friend instances after SPA navigation.
- Handle browser zoom and window resizing.
- Handle fullscreen video.
- Avoid covering important controls when possible; allow user repositioning.
- Gracefully fail on pages where content scripts cannot run, such as Chrome internal pages.

19. Testing Plan

Area	Test cases
Browsers	Current Chrome stable; later Chromium-based browsers if desired.
Websites	YouTube, GitHub, Google, Reddit, news sites, ordinary HTML pages.
Video	Play, pause, seek, end, fullscreen, switching videos.
Layout	Scrolling, resizing, browser zoom 80–125%+, different monitor sizes.
Interaction	Click, hover, keyboard shortcut, popup controls.
Privacy	Toggle Privacy Mode repeatedly; reload pages; restart Chrome; verify persistence.
Performance	Idle browser, active video, long session, multiple tabs.
Failure	Missing assets, malformed settings, SPA navigation, DOM replacement.

20. Security & Privacy Strategy

- Keep V1 entirely client-side.
- Do not collect browsing history.
- Do not send page contents or URLs to a server.
- Avoid analytics unless there is a strong reason and the privacy implications are fully understood.
- Request only the permissions actually needed.
- Document exactly what the extension can access and why.
- If a backend is introduced later, redesign the privacy model rather than silently adding data collection.

21. Development Milestones

Milestone	Deliverable	Definition of done
M1	Extension skeleton	Loads unpacked and opens popup.
M2	Static friend	Friend appears reliably on test pages.
M3	Animation engine	Frame and sprite animations work.
M4	State machine	Autonomous activity transitions work.

Milestone	Deliverable	Definition of done
M5	Core activities	Idle, sitting, eating, sleeping, shower, watching work.
M6	Browser awareness	Generic video + YouTube behavior works.
M7	Controls	Size, position, hide, sound, interaction controls work.
M8	Privacy Mode	Shortcut + persistent focus mode work.
M9	Optimization	Long sessions do not create obvious performance problems.
M10	Release candidate	Store-ready package and documentation.

22. Recommended Build Order

01. Create empty project and Manifest V3.
02. Load unpacked extension in Chrome.
03. Implement popup shell.
04. Inject a fixed overlay with Shadow DOM.
05. Display one placeholder character.
06. Implement size and position controls.
07. Build the frame/sprite animation engine.
08. Create the idle animation.
09. Implement the state machine.
10. Add sitting, eating, sleeping, shower, watching, looking, and reaction states.
11. Implement randomized autonomous behavior.
12. Add time-of-day behavior.
13. Add generic HTML5 video detection.
14. Add YouTube-specific detection.
15. Implement click/hover interactions.
16. Implement Hide.
17. Implement Privacy Mode and keyboard shortcut.
18. Persist settings.
19. Optimize assets and runtime behavior.
20. Test across sites, resolutions, zoom levels, and long sessions.
21. Prepare icons, screenshots, listing copy, and privacy documentation.
22. Package as ZIP.
23. Create Chrome Web Store developer account and submit.
24. Respond to review feedback if required.
25. Publish V1.0.

23. Future Roadmap

Version	Potential additions
1.1	More animations, more reactions, timed hiding, better site adapters.
1.5	Personality traits, mood, energy, familiarity/relationship memory, richer routines.
2.0	Multiple characters, deeper browser context, optional conversational features.
2.x	Potential Live2D or more advanced animation if the visual style needs it.
Desktop edition	Separate Electron/Tauri/native overlay if the friend should appear outside Chrome.

24. What NOT to Build Yet

- Do not build a backend just because the extension is going public.
- Do not add an AI chat model before the autonomous visual behavior feels good.
- Do not start with Live2D unless the art style absolutely requires it.
- Do not implement every website separately.
- Do not add dozens of hidden stats that don't visibly affect behavior.
- Do not optimize for every Chromium browser before Chrome V1 is stable.

25. Production Checklist

- Manifest V3 validated
- No unnecessary permissions
- No console errors during normal use
- No duplicate friend instances
- Shadow DOM isolation verified
- Animations work after navigation
- Video behavior works
- Privacy Mode works
- Keyboard shortcut works
- Settings persist after restart
- Missing assets fail gracefully
- Performance checked during long sessions
- Store icon and screenshots prepared
- Store description prepared
- Privacy disclosures prepared
- ZIP contains manifest.json at its root
- Version number incremented correctly
- Release candidate tested from a clean Chrome profile

26. Publishing & Cost

- **Local development:** no hosting cost is required. You can load the unpacked extension through Chrome's Extensions page.
- **Backend:** not required for the planned V1 because the friend, assets, behavior, and settings can run locally.
- **Chrome Web Store:** publishing requires a developer registration and the store's current account requirements. Google documentation confirms that developers must enable 2-step verification to publish/update items and must complete the Store listing and Privacy tabs before publishing. ■cite■turn0search0■
- **Current publishing process:** package the extension, upload it to the Developer Dashboard, complete listing/privacy information, and submit it for review. Google's 2026 documentation says publishing submits the item for review. ■cite■turn0search0■
- **Important:** verify the current registration fee directly in the Chrome Web Store Developer Dashboard when you are ready to publish; fees and account requirements can change.

27. Suggested V1 Definition of Done

V1 is complete when a fresh Chrome profile can install the unpacked extension, open normal webpages, see a small virtual friend in a corner, watch the friend autonomously move through believable activities, detect video playback, react occasionally, allow size/position controls, and instantly disappear through Privacy Mode. The extension should remain local-first, lightweight, and stable during a long browsing session.

28. Key Architectural Rule

Behavior → State → Animation → Rendering

Do not tie personality logic directly to individual image files. The behavior engine should request an activity such as WATCHING or SLEEPING. The animation layer decides which frames to display. The renderer decides where and how large the friend appears. This separation makes future character packs, new animation technologies, and personality systems much easier to add.

Reference note: Chrome Web Store and Chrome extension requirements are based on current Google Chrome for Developers documentation available in September 2026. ■cite■turn0search0■turn0search3■